

What Biopsychosocial Factors Impact Running During the First Year Postpartum.

Emily Fitzwilliam¹, Danielle Trowell¹, Jason Bonacci¹, Samantha Hoffmann, Aaron Fox¹ and Natalie Saunders¹

¹ Institute for Physical Activity and Nutrition, Deakin University

Email: s224769019@deakin.edu.au

INTRODUCTION

Physical activity (PA) is a therapeutic modality that can support postpartum (PP) recovery and optimise overall maternal health [1]. PP PA engagement is reduced, due to time, energy, and healing demands [1], increasing the risk of serious health consequences [2]. Moreover, they frequently do not return to pre-pregnancy PA levels [3], even 3 years PP [4]. Running is popular amongst mothers, largely due to its adaptability to time and cost constraints [1]. Regular running mitigates more than 26 conditions [5], without negatively impacting breastfeeding [6]. Despite these advantages, at almost 1 year PP, 63% of women who are running have not returned to their pre-pregnancy level, compromising running adherence to optimise health outcomes [7]. Many women feel unsupported by health professionals during this journey [8]. However, due to a lack of evidence-informed guidelines and at times, conflicting evidence, the ability for clinicians to provide consistent guidance for individualised care is reduced. Thus, this study aims to determine which BPS factors impact running and how during the first twelve weeks of running initiation, within the first year PP.

METHODS

Deakin University Human Research Ethics Committee approved this project (2025/HE000695). Participants were recruited from a preceeding investigation, exploring preparation to running PP. This facilitated a known point of progression into continuous running (i.e., >10 minutes in a session, requirement for this study), regardless of weeks PP. Participants completed an online survey, each fortnight across 12 weeks. 27 factors were included in the survey across biological, psychological, social and other domains. For each BPS factor, subjects were asked if the factor impacted running across the previous fortnight, if not, they had to identify if this was as it did not exist (0), or that it had no impact on running (0). If the factor did impact running, however, participants had to rate its impact on a Likert scale, ranging from strongly (-2) and somewhat (-1) negative, to somewhat (+1) and strongly (+2) positive. Participants had to provide at least two survey responses to be included. A descriptive frequency analysis was performed for each BPS factor, calculating the percentage of responses falling into each of the six categories. The weighted impact strength was calculated using the mean positive (i.e., +1 and +2) and negative (i.e., -1 and -2) strength scores for responses, multiplied by the proportion of the sample in that category.

RESULTS AND DISCUSSION

82 participants (Age: 34 ± 5 years; Weight: 74 ± 14 kg; Height: 166 ± 7 cm; Postpartum: 44 ± 15 weeks) contributed 339 responses. The percentage of responses

in each category for the BPS factors and the associated weighted mean is displayed in Figure 1. The BPS factors that frequently had a negative and strongly rated impact on running were motherhood responsibilities (69.6% negative, -0.95 strength), time (51.6% negative, -0.67 strength), and energy (50.7% negative, -0.67 strength). The BPS factors that frequently had a positive and strongly rated impact on running were support from a partner (34.8% positive, +0.58 strength) and friends (15% positive, +0.24 strength), and motivaion (14.7% positive, +0.22 strength). Whilst factors such as time and energy corroborate prior suggestions [9, 10], others such as partner and friend support have not been previously highlighted. Motherhood responsibilities and time demands are largely unmodifiable, promoting individualised running strategies to overcome constraints [11]. To help navigate this, support from a partner or friends, should be made tangible through involvement in goal setting, planning, preparation such as organising the baby and pram, childcare and runing itself.

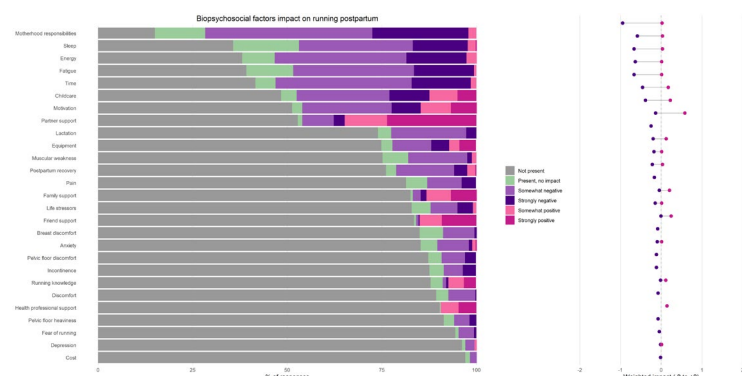


Figure 1. Stacked barchart of frequency analysis and weighted strength for biopsychosocial factors impacting running postpartum.

CONCLUSIONS

Motherhood responsibilities, time and energy, and then partner and friend support, and motivation were the strongest barriers and enablers respectively. Practical ways to overcome barriers and promote enablers to aid running adherence should be the focus of future work.

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